

MOST REPEATED POWER BI INTERVIEW QUESTIONS AND ANSWERS



1. How do you use Power BI's Q&A feature?

The Q&A feature allows users to ask natural language questions about their data, and Power BI generates instant visualizations or metrics as answers.

For example, you can type "Show total sales by region" or "What is the profit trend over time?" to get results. It's important to ensure the dataset is well-prepared with clear names and relationships. You can also train the Q&A model by adding synonyms or refining terms in Q&A settings. Once the visuals are created, they can be pinned to dashboards or shared with stakeholders.

2. How do you handle data refresh issues in Power BI?

To handle data refresh issues, I first check the error message and refresh history to identify the cause.

Common fixes include updating credentials, resolving gateway issues, or aligning Power Query steps with schema changes. For performance problems, I optimize queries to reduce data volume and check timeout settings. For scheduled refresh failures, I ensure all linked datasets are refreshed in order and monitor them with alerts. If the issue persists, I escalate it to the IT team or Power BI support.

3. How do you create a calculated table in Power BI?

To create a calculated table, I use DAX in the "Modeling" tab by selecting New Table. Calculated tables are derived from existing data and can be used to create relationships, summarize data, or generate dynamic tables.

For example, I can create a table with summarized sales by region using:
`RegionSales = SUMMARIZE(Orders, Orders[Region], "Total Sales", SUM(Orders[Sales]))`.

Calculated tables are updated dynamically with the data model.

4. How do you handle large datasets in Power BI?

To handle large datasets, I enable DirectQuery or Live Connection instead of importing data to limit memory usage. I also optimize data by removing unused columns, filtering rows, and aggregating data in the source itself. Additionally, I use partitioning and incremental refresh for time-based data and ensure efficient query performance by leveraging database indexing or views.

5. Describe how to use Power BI with Azure services.

Power BI integrates seamlessly with Azure services like Azure SQL Database, Data Lake, and Synapse Analytics. For instance, I can connect Power BI to Azure SQL for real-time reporting or use Azure Data Lake for large-scale data storage. With Azure Synapse, I leverage its analytics capabilities to process and visualize data efficiently in Power BI. Azure Active Directory is used for secure authentication.

6. How will you join two tables in Power Query?

In Power Query, I join two tables using the Merge Queries option. After selecting the tables, I choose the common column to define the relationship. Power Query provides different join types like Inner, Left Outer, and Full Outer joins. Once merged, I can expand the resulting table to include the desired columns. This is useful for combining related data sources.

7. You are working with a large dataset that causes Power BI to crash when loading. What techniques would you use to handle and analyze the data efficiently?

I would first reduce the dataset size by applying filters or aggregating data at the source. Switching to DirectQuery or Live Connection helps avoid importing large data. Using incremental refresh minimizes data load by only updating recent data. I also optimize the model by removing unused columns and rows and ensuring all queries are efficient. For very large datasets, I leverage tools like Azure Synapse or dataflows.

8. What is a Power BI dataflow, and how is it used?

A Power BI dataflow is a self-service ETL (Extract, Transform, Load) tool that allows you to prepare, transform, and load data into Power BI using Power Query in the cloud. It is used to centralize data preparation, enabling reusability across multiple reports and datasets. Dataflows are often stored in Azure Data Lake, allowing integration with

other services. They help ensure consistency in data transformation and reduce redundancy in data preparation.

9. How do you integrate Power BI with other Microsoft services?

Power BI integrates seamlessly with services like Excel, SharePoint, Teams, and Azure. For example, I can connect Power BI to SharePoint lists or Excel files stored in OneDrive for dynamic updates. It works with Teams for sharing reports and collaborating in real-time. Azure services like SQL Database and Synapse Analytics provide scalable data storage and analysis, and Power BI integrates with Microsoft Power Automate to create workflows for report notifications or actions.

10. What are the key components of a Power BI report?

Key components of a Power BI report include:

- Visualizations: Charts, graphs, and tables that represent data insights.
- Filters: Used to narrow down data based on user selections.
- Slicers: Visual controls to interactively filter data.
- KPIs: Metrics that display progress against goals.
- Report Pages: Multiple pages within a report for organizing visuals.
- Drillthrough and Drill Down: Allow deeper exploration of data by navigating between levels of detail.

11. How do you secure sensitive data in Power BI reports?

To secure sensitive data, I implement Row-Level Security (RLS) to restrict access based on user roles. I also ensure sensitive columns are masked or removed from reports and leverage encryption for datasets in the cloud. In the Power BI Service, I restrict sharing permissions, disable export options like Excel and CSV, and configure workspace security settings to control access.

12. You encounter a data integrity issue where some records are duplicated in the dataset. How would you resolve this in Power BI?

To resolve issues, I first identify the duplicate records using Power Query by applying the Group By operation or filtering for duplicates. Then, I remove duplicates using the Remove Duplicates option on the relevant column(s). If duplicates are intentional but need to be handled, I use aggregation functions or create a DAX measure to account for them in calculations.

13. What is RANK and DENSE_RANK in SQL?

Both RANK() and DENSE_RANK() are window functions in SQL that assign a rank to rows based on a specified column or expression.

- RANK(): Leaves gaps in the ranking if there are ties (e.g., ranks 1, 2, 2, 4).
- DENSE_RANK(): Does not leave gaps and assigns consecutive ranks (e.g., ranks 1, 2, 2, 3).

They are often used in analytical queries to rank items like sales performance or customer orders.

14. Suppose there is a large query with multiple joins and logic that used to execute in 10–15 minutes but now takes over 30 minutes. How would you diagnose and resolve this?

- Analyze the execution plan using EXPLAIN to identify slow steps.
- Check for missing or outdated indexes on WHERE, JOIN, and ORDER BY columns.
- Investigate table size growth or recent query logic changes.
- Look for blocking or deadlocks and ensure database statistics are updated.
- Optimize indexes, simplify query logic, or consider partitioning/archiving for large tables.

15. Can you explain the types of indexes in SQL and how they work?

- Clustered Index: Physically sorts table rows; one per table.
- Non-Clustered Index: Creates pointers to data; multiple allowed per table.
- Unique Index: Ensures column values are unique.
- Full-Text Index: Optimized for searching text-based data.
- Composite Index: Combines multiple columns into a single index.

16. What are DDL, DML, and DQL commands in SQL?

- DDL (Data Definition Language): Commands like CREATE, ALTER, DROP to define structure.
- DML (Data Manipulation Language): Commands like INSERT, UPDATE, DELETE for data handling.
- DQL (Data Query Language): Command SELECT to retrieve data from tables.

17. How would you find the 5th highest salary in an employee table?

```
SELECT DISTINCT salary  
FROM employees
```

ORDER BY salary DESC

LIMIT 1 OFFSET 4;

Use LIMIT with OFFSET to skip top 4 salaries and fetch the 5th.

18. What is the difference between UNION and UNION ALL in SQL?

UNION combines the results of two queries and removes duplicates, while UNION ALL includes all records, even if they are duplicates. UNION is useful when you need unique records, but it has extra processing overhead due to duplicate elimination. UNION ALL is faster because it does not perform this check, making it a better choice when duplicates are acceptable.

19. How do you calculate the years an employee has spent in the organization using DAX in Power BI?

YearsSpent = DATEDIFF(Employee[StartDate], TODAY(), YEAR)

This calculates the difference between StartDate and today in years.

20. Suppose you have two fact tables, Fact A and Fact B, and three dimension tables, Dimension A, Dimension B, and Dimension C. Fact A is connected to all three dimension tables, and Fact B is also connected to the same three dimension tables. How can you establish a relationship between Fact A and Fact B?

- Create a bridge table or shared dimension with keys from both fact tables.
- Use the shared dimension to establish indirect relationships between the facts.
- Alternatively, use DAX to calculate measures that link data from both fact tables.
- Avoid direct relationships between fact tables to prevent ambiguity.

21. What is the difference between Table and Matrix in Power BI?

- Table: Displays data in a simple tabular format with rows and columns; no aggregation is applied.
- Matrix: Functions like a pivot table, allowing aggregation and hierarchical views for row and column data.
- Matrix supports cross-highlighting and better visualization for summarized data.

22. If I have two tables in Power BI with a many-to-many relationship, how can I avoid it?

- Introduce a bridge table with distinct keys from both tables to act as a mediator.

- Redesign the data model to eliminate duplicates in key columns.
- Use summarization or aggregation to reduce granularity and avoid direct many-to-many links.

23. Explain the scenario and provide the best-optimized solution: You have 100 Excel files, each representing daily sales or order data.

- Use Power Query to combine all files by connecting to the folder containing them.
- Power Query automates the transformation process and consolidates data into a single dataset.
- Ensure that all files have consistent column headers for accurate data merging.

24. How many active relationships are allowed between two tables in Power BI?

- Power BI allows only one active relationship between two tables.
- You can define multiple inactive relationships and activate them in calculations using USERELATIONSHIP.

25. Why would you use the DATEADD function in Power BI?

- The DATEADD function shifts dates by a specified interval, such as days, months, or years.
- It is commonly used for time intelligence calculations like year-over-year or month-over-month comparisons.

26. What is the difference between DirectQuery and Live Connection?

- DirectQuery: Queries data directly from the data source and allows custom modeling and DAX calculations.
- Live Connection: Connects to a pre-existing model (e.g., SSAS, Power BI Dataset) without additional modeling or transformations.
- DirectQuery supports multiple data sources, while Live Connection is limited to one.

27. Are you aware of Microsoft Fabric?

- Yes, Microsoft Fabric is an integrated analytics platform that unifies data engineering, data science, and BI tools.
- It combines services like Power BI, Synapse, and Data Factory to provide end-to-end analytics workflows.

28. What major challenges have you faced in your Power BI projects?

- Optimizing performance for large datasets using DirectQuery.
- Managing complex data models and ensuring accurate calculations in measures.

- Configuring gateways for on-premises data sources and automating refresh schedules.
- Implementing and testing Row-Level Security (RLS) for restricted user access.

29. Did you have access to a premium Power BI account?

Yes/No (based on your experience). I am familiar with the features of Power BI Premium, such as:

1. Enhanced Performance and Capacity: Provides dedicated capacity for faster report loading and dataset refresh.
2. AI Capabilities: Access to AI features like AutoML, cognitive services, and dataflows for more advanced analytics.
3. Paginated Reports: Ability to create pixel-perfect paginated reports suitable for printing.
4. Larger Dataset Sizes: Premium supports datasets up to 400 GB.
5. Deployment Pipelines: A streamlined process for managing development, testing, and production environments.

If required, I can adapt quickly to a Premium environment and leverage its features to handle large-scale enterprise reporting needs.

30) Types of Analysis

1. Descriptive Analysis:
 - Focuses on summarizing past data to answer 'What happened?'
 - Example: Analyzing historical sales data to identify trends.
2. Diagnostic Analysis:
 - Explores the reasons behind events to answer 'Why did it happen?'
 - Example: Investigating why sales dropped in a particular quarter by analyzing regions, products, or campaigns.
3. Predictive Analysis:
 - Uses statistical models and machine learning to forecast future outcomes and answer 'What is likely to happen?'
 - Example: Predicting next quarter's sales based on historical trends.
4. Prescriptive Analysis:
 - Provides recommendations and insights to answer 'What should we do?'
 - Example: Suggesting the best pricing strategy for maximizing profit.
5. Exploratory Analysis:
 - Focuses on uncovering patterns or relationships within data when there's no predefined question.
 - Example: Identifying potential customer segments from a large dataset.

6. Inferential Analysis:

- Involves making predictions or generalizations about a population based on a sample.
- Example: Using survey data to infer customer satisfaction for the entire customer base.

31. Difference Between Data Lake, Data Warehouse, and Data Mart

Feature	Data Lake	Data Warehouse	Data Mart
Purpose	Stores raw, unstructured, or semi-structured data for large-scale analytics.	Stores structured data optimized for reporting and analysis.	A subset of a data warehouse, focused on specific business areas.
Data Type	Raw and unprocessed data.	Processed and structured data.	Highly curated and structured data.
Use Case	Data exploration, big data analytics, and machine learning.	Business intelligence and operational reporting.	Department-specific analytics, e.g., sales or marketing.
Storage Cost	Cost-effective due to cheap storage solutions (e.g., Azure Blob Storage, S3).	Higher storage costs because data is optimized.	Similar to Data Warehouse but smaller in scale.
Technology	Hadoop, Azure Data Lake, Amazon S3, etc.	Snowflake, Redshift, SQL Server, etc.	Part of a warehouse, created using the same technologies.
Accessibility	Designed for data scientists and engineers.	Designed for analysts and business users.	Designed for department-specific users.
Schema	Schema-on-read (defined during analysis).	Schema-on-write (defined before data ingestion).	Schema-on-write.

Key Differences:

1. A Data Lake is for raw data storage and exploratory analysis.
2. A Data Warehouse is for structured and processed data, ideal for reporting.
3. A Data Mart is a focused subset of a Data Warehouse for a specific business function.

32. Difference Between Structured and Unstructured Data

Structured Data:

- Organized into predefined formats like rows and columns.
- Easy to search, query, and analyze using SQL and BI tools.

Examples: Sales data, customer databases, transaction logs.

Unstructured Data:

- Doesn't have a predefined structure, making it harder to analyze.
- Requires specialized tools for processing and analysis.
- Examples: Text files, videos, images, emails, social media posts.

33. Difference Between OLTP and OLAP

Feature	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)
Purpose	Manages transactional data for day-to-day operations.	Analyzes data for decision-making and reporting.
Data Type	Real-time, current data.	Historical and aggregated data.
Operations	Insert, Update, Delete.	Read-intensive (e.g., queries, reports).
Examples	Banking systems, e-commerce platforms.	Dashboards, sales analysis, forecasting tools.
Performance	Optimized for fast transactional updates.	Optimized for complex queries and large datasets.

34. Difference Between Type 1 and Type 2 Slowly Changing Dimensions (SCDs)

Type 1 SCD:

- Overwrites the old data with new data, maintaining only the current state.
- No historical tracking.
- Simple to implement.
- Use Case: When historical data isn't important, like correcting typos in names.

Type 2 SCD:

- Preserves history by adding a new record for every data change.
- Tracks changes over time with additional metadata (e.g., Start Date, End Date).
- More complex to implement but provides a complete history of changes.
- Use Case: Tracking customer address changes over time.

35. Difference between Delete, Drop & Truncate?

Feature	DELETE	TRUNCATE	DROP
Type	DML	DDL	DDL
Removes	Specific rows (with WHERE) or all rows	All rows in the table	Entire table (data + structure)
Condition Support	Yes (with WHERE)	No	No
Transaction Control	Can be rolled back	Cannot be rolled back	Cannot be rolled back
Triggers	Yes (fires triggers)	No	No
Performance	Slower for large datasets	Faster for large datasets	N/A (removes entire table)
Table Structure	Retained	Retained	Deleted

36. How to Remove Duplicate Records from a Table

To remove duplicate rows from a table, you can use a DELETE statement with a Common Table Expression (CTE) or a ROW_NUMBER() function in SQL. Here's how:

WITH CTE AS (

SELECT

*,

ROW_NUMBER() OVER (PARTITION BY column1, column2, column3 ORDER BY id) AS row_num

FROM table_name

)

DELETE FROM table_name

WHERE id IN (

SELECT id FROM CTE WHERE row_num > 1

);

-

Explanation:

- The ROW_NUMBER() function assigns a unique number to each row within a group of duplicates based on columns (column1, column2, etc.).
- Rows with row_num > 1 are duplicates and are deleted.

37. Difference Between UNION and UNION ALL

Feature	UNION	UNION ALL
Purpose	Combines results from two queries and removes duplicates.	Combines results from two queries without removing duplicates.
Duplicates	Removes duplicates automatically.	Keeps all duplicates.
Performance	Slower (due to duplicate elimination).	Faster (no duplicate check).
Usage	Use when duplicate rows are not needed.	Use when duplicates are acceptable or necessary.
Syntax	SELECT ... UNION SELECT ...;	SELECT ... UNION ALL SELECT ...;

38. What is a Power BI dataflow, and how is it used?

A dataflow is a collection of ETL (Extract, Transform, Load) processes in Power BI, used to clean, transform, and prepare data for use in datasets and reports.

Usage:

Centralize data preparation.

Reuse data across multiple datasets.

Perform data transformations using Power Query.

39. How do you integrate Power BI with other Microsoft services?

Excel: Import or export data.

SharePoint: Embed reports or fetch data.

Teams: Share and collaborate on reports.

Azure: Connect with Azure SQL, Synapse, or Data Lake.

Power Automate: Automate workflows based on Power BI triggers.

40. How do you set up data alerts in Power BI?

Create a KPI, card, or gauge visual in your report.

Publish the report to Power BI Service.

Set up alerts by clicking the visual's ellipsis (...) and selecting "Manage alerts."

Define conditions for the alert (e.g., threshold values).

Enable email notifications.

41. How do you secure sensitive data in Power BI reports?

Row-Level Security (RLS): Restrict access to data based on user roles.

Permissions: Use workspace roles to control access.

Masking: Use DAX to hide sensitive details.

Encryption: Ensure data is encrypted in transit and at rest.

42. You encounter a data integrity issue where some records are duplicated in the dataset. How would you resolve this in Power BI?

Use Power Query to:

Remove duplicates using the "Remove Duplicates" feature.

Filter rows based on conditions.

Use DAX to identify duplicates:

Duplicate Count = COUNTROWS(FILTER(table_name, COUNTROWS(table_name) >

1))

43. How does the end user access the report?

Through Power BI Service via shared links or apps.

Embedded in web portals or applications.

Shared via Microsoft Teams or email.

Exported as PDF or PowerPoint presentations.

44. How do you deploy reports from one environment to another?

To deploy reports between environments (e.g., Dev to Test to Prod), I use the following steps:

Ensure the dataset and report are published to a shared workspace.

Parameterize connections (e.g., server/database names).

Use deployment pipelines for versioning and tracking changes.

Validate data connections in the target environment.

Perform quality assurance checks.

45. Can you explain workspace management in Power BI?

Workspace management involves:

Creating and assigning roles (Admin, Member, Contributor, Viewer).

Organizing datasets, reports, dashboards, and dataflows within a workspace.

Setting access permissions to ensure data security.

Managing scheduled refreshes and resource capacities.

46. What are the different versions of Power BI?

Power BI Desktop: For creating reports and dashboards.

Power BI Service: For publishing, sharing, and collaboration.

Power BI Pro: Enables sharing and collaboration with others.

Power BI Premium: Offers dedicated capacity and additional features like paginated reports.

Power BI Mobile: For accessing reports on the go.

47. Have you faced data type errors? Can you share an experience?

Yes, once I encountered a mismatch between date formats in the dataset and the source. This caused aggregation errors. I resolved it by:

Transforming the column to a uniform date format in Power Query.

Validating with the source system for consistent formatting.

48. Have you used Power Automate and Power Apps?

Yes, I used Power Automate for automating workflows like sending email alerts for report updates. I used Power Apps to create custom forms for capturing user inputs that directly integrate with Power BI datasets.

49. How do you gather requirements from clients?

Start with stakeholder meetings to understand business goals.

Ask specific questions about KPIs, data sources, and expected visualizations.

Create wireframes/mockups for initial approval.

Ensure continuous feedback during development.

50. What types of transformations have you applied in your reports or data models?

In my Power BI projects, I've used various transformations, including:

Data Cleansing: Removing duplicates, handling null values, and fixing data types.

Column Transformations: Merging, splitting, and creating calculated columns.

Aggregations: Summarizing data for higher-level insights.

Unpivot: Converting columns into rows for easier analysis.

Conditional Columns: To apply business rules and classifications.

51: Can you name some important KPIs for the insurance and sales domains?

Insurance Domain:

Claim Frequency and Severity

Loss Ratio

Customer Retention Rate

Incurred but Not Reported (IBNR) Claims

Sales Domain:

Total Sales (Sales – Returns)

% Target Achieved

Customer Lifetime Value

Profit Margin %

52: What was the size of your data model?

My largest data model had approximately 2 million rows across multiple tables, optimized using star schema and aggregations for performance.

53: Which visuals did you use and why?

I've used:

Bar/Column Charts: For trend analysis and category comparison.

Line Charts: To show sales trends over time.

Pie/Donut Charts: For contribution analysis (e.g., market share).

Tables: For detailed tabular data insights.

KPIs: To track performance against targets.

54: What's the difference between DATEPERIOD and DATEADD functions in Power BI?

DATEPERIOD: Creates a range of dates based on specific intervals like year, quarter, or month.

DATEADD: Shifts dates by a specified interval, e.g., moving forward/backward by days, months, or years.

Example: Comparing current and previous period sales.

55: What are Pivot and Unpivot? Provide examples.

Pivot: Converts rows into columns (e.g., converting product categories into separate columns).

Unpivot: Converts columns into rows (e.g., transforming monthly sales columns into two rows: Month and Sales).

56: Difference between a view and a stored procedure, with examples.

View: A virtual table representing a SELECT query.

Example:

```
CREATE VIEW SalesView AS  
  
SELECT Product, SUM(Sales) AS TotalSales  
  
FROM Orders  
  
GROUP BY Product;
```

Stored Procedure: A saved set of SQL queries that accepts parameters. Example:

```
CREATE PROCEDURE GetSales (@ProductID INT)  
  
AS  
  
SELECT * FROM Orders WHERE ProductID = @ProductID;  
  
EXEC GetSales 101;
```

57.What is the difference between a reference query and an SQL query in Power BI?

Reference Query: Creates a new query linked to an existing one, inheriting all transformations. Ideal for creating variations of the same dataset without duplication.

SQL Query: Executes a pre-written SQL script against the database to retrieve data directly.

58. Difference between Power BI Pro and Power BI Premium licenses.

Power BI Pro: For individual users; includes collaboration, sharing, and 1 GB dataset size limit.

Power BI Premium: For organizations; includes larger dataset sizes, paginated reports, AI features, and on-premises reporting.

59: Explain the SUMMARIZE DAX function with an example.

SUMMARIZE creates a summary table with specified groupings.

```
SUMMARIZE(Sales, Sales[Region], "Total Sales", SUM(Sales[Amount]))
```

This groups data by Region and calculates Total Sales for each group.

60: What types of reports have you created in Power BI?

Executive Dashboards: KPIs, trends, and performance metrics.

Operational Reports: Detailed daily or transactional data.

Paginated Reports: Printable reports for invoices and financials.

Custom Visualizations: Focused on user-specific insights, such as sales vs. targets.

61: Have you worked with paginated reports? How do they differ from standard reports?

Yes! Paginated reports are pixel-perfect, designed for printing or export, suitable for detailed, page-oriented data (e.g., invoices). Standard reports are interactive and optimized for on-screen analysis.

62: How do you create dynamic titles in Power BI reports?

By using DAX measures like:

DynamicTitle = "Sales Report for " & SELECTEDVALUE(Region[Region])

Apply this measure in a card visual to display a title based on filters.

63: What is DAX Studio, and how can it help optimize queries?

DAX Studio is a tool for analyzing and optimizing DAX queries. It helps debug, check performance, and identify bottlenecks using query plans and metrics.

64: What is more important: "Accurate Numbers" or "Best Visualization"?

Answer: Both are critical, but accuracy takes precedence. Without accurate data, even the best visualizations mislead decision-makers. However, combining accuracy with effective visuals provides maximum insights.

65: Have you created BRD/Closure/New Release/Testing documents?

Yes, I've worked on:

BRD (Business Requirement Document): Captures project requirements.

Closure Docs: Summarizes project completion.

New Release Docs: Documents feature updates.

Testing Docs: Details test cases and results.

66) How do you use bookmarks in Power BI to enhance report interactivity?

Bookmarks capture the current state of a report page, including filters, slicers, visibility of objects, and drill-through states. They can be used to create navigation, toggle between different views, or enhance storytelling by allowing users to switch between customized report layouts.

67) How do you implement row-level security (RLS) in Power BI?

Use roles in Power BI Desktop to define filters that restrict data at the row level. Then, publish the report to Power BI Service and assign roles to specific users.

68) What are the best practices for designing a Power BI dashboard?

Focus on clarity, use a consistent color scheme, keep visualizations simple, ensure data accuracy, and tailor the dashboard to the end user's needs.

69) Can you explain the Power BI report lifecycle?

It includes development in Power BI Desktop, publishing to Power BI Service, sharing with users, and maintaining through regular updates and data refreshes.

70) How does Power BI handle data refresh and scheduling?

Power BI allows scheduled refreshes based on the dataset, which can be configured in Power BI Service. Direct Query and Live Connection also enable near real-time data updates.

71) What is the role of the Query Editor in Power BI?

The Query Editor is used for data transformation tasks such as filtering, cleaning, and reshaping data before loading it into the Power BI model.

72) How do you create a funnel chart in Power BI?

Use the Funnel Chart visual, typically to represent a process that has stages, such as a sales pipeline, showing progressive reduction.

73) What is the difference between a measure and a KPI in Power BI?

A measure is a calculation used in data analysis, while a KPI (Key Performance Indicator) is a visual representation that tracks performance against a target.

74) Explain the concept of hierarchical slicers in Power BI.

Hierarchical slicers allow users to drill down into multiple levels of related data, providing a more detailed filtering experience.

75) How can you create and configure a data gateway in Power BI?

Step 1: Download the gateway from the Power BI website and install it on a server in your network.

Step 2: Sign in with your Power BI account.

Step 3: Configure Gateway: Name your gateway and set recovery keys.

Step 4: Connect to Your Data Source: In Power BI Service, go to settings, choose "Manage Gateways," and add your data sources (e.g., SQL Server, SharePoint).

Step 5: Assign Gateway to Datasets: Link your Power BI datasets to the configured gateway for scheduled refreshes and live queries.

76) How does Power BI handle real-time data streaming, and what are its use cases?

Step 1: Set Up Streaming Dataset: In Power BI Service, go to the "Datasets" section, click on "Create," and choose "Streaming dataset."

Step 2: Choose Data Source: Select from different streaming options like API, Azure Stream Analytics, or PubNub.

Step 3: Define Data Fields: Specify the data fields that will be included in the stream,

such as sensor readings, timestamps.

Step 4: Build Dashboard: Use the streaming dataset to build real-time visuals on a dashboard, such as line charts, gauges, or cards that update instantly as new data arrives.

Use Cases: Real-time data streaming is ideal for monitoring IoT devices, tracking live events, financial transactions, or any scenario where immediate insights are crucial.

77) Describe the purpose and usage of Power BI Premium.

Power BI Premium offers dedicated capacity for better performance, large datasets, and advanced features like paginated reports. It's suited for organizations needing extensive data processing and sharing capabilities.

78) How do you create a sunburst chart in Power BI?

Although not native to Power BI, sunburst charts can be created using custom visuals. They visually represent hierarchical data, such as organizational structures or product categories.

79) What are the differences between a line chart and an area chart in Power BI?

- **Line charts** display trends over time with clear lines, making it easy to see fluctuations.
- **Area charts** fill the space below the lines, emphasizing volume and magnitude of change, which is useful for showing cumulative trends.

80) What is column profiling?

Column Profiling: This process involves examining data columns for statistics such as data types, value distribution, and missing data, ensuring data quality before analysis.

81) How will you join two tables with the help of a DAX function?

Joining Tables with DAX: Use the **RELATED** function to fetch data from related tables in one-to-many relationships. Alternatively, **LOOKUPVALUE** can be used for more complex scenarios.

82) What is cross-filter?

Cross-Filter: This feature enables interactivity between visuals on a report. Selecting data in one visual automatically filters data in related visuals, enhancing the analytical experience.

83) How will you refresh a single table in Power BI?

Refreshing a Table: In Power BI Desktop, you can refresh a specific table by right-

clicking the table and selecting "**Refresh Data.**" In Power BI Service, you would refresh the dataset, which updates all tables.

84) Which is preferred: slicer or filters?

Slicers vs. Filters:

- **Slicers** are more intuitive for end-users and provide a visual way to filter data.
- **Filters** offer more control, especially for applying conditions across multiple visuals or pages.

85) How do you give permission to 50 users in Power BI?

Granting Permissions: In Power BI Service, you can add users to a **workspace** or share reports directly with them by email. For larger groups, consider using **security groups** or **distribution lists**.

86) How do you apply the same slicer across multiple pages?

Syncing Slicers: In Power BI, use the **Sync Slicers** feature to apply the same slicer across multiple report pages, ensuring consistent filtering.

87) Difference between galaxy and star schema?

- **Star Schema:** A single central fact table linked to dimension tables, leading to simpler queries and better performance.
- **Galaxy Schema (Snowflake Schema):** Involves multiple fact tables and normalized dimension tables, making it more complex but reducing data redundancy.

88) What are field parameters?

Field Parameters: These allow users to dynamically change the fields used in a visual, making reports more interactive and customizable based on user selection.

89). How do you create a Pareto chart in Power BI?

Creating a Pareto Chart: Combine a column chart and a line chart on the same visual, where the columns represent individual values sorted in descending order, and the line represents the cumulative percentage of the total.

90). What are the advantages of using Power BI templates?

Power BI Templates: Templates (.pbix files) allow users to save the structure of a report, including visuals, themes, and queries, without the data. This is ideal for creating standardized reports that can be reused across different datasets.

91) What are Power BI datasets, and how do they differ from data sources?

Datasets: Processed and structured data stored within Power BI, ready for analysis. Data sources are the raw origins of the data, like databases or Excel files.

92) How do you handle data refresh issues in Power BI?

Data Refresh: Check gateway status, data source credentials, and scheduled refresh settings. Ensure the data model is optimized to avoid timeouts.

93) Using Power BI Embedded for application development?

Power BI Embedded allows embedding Power BI reports into custom applications via APIs, enabling users to interact with reports without needing a Power BI account.

94) What is the distinction between DirectQuery and Live Connection in Power BI?

- **DirectQuery:** Retrieves data from the source on demand, enabling near real-time analysis. Supports multiple data sources.
- **Live Connection:** Connects to an existing model in SSAS without importing data. Limited to a single model.
- **Key Difference:** DirectQuery queries the source directly, while Live Connection relies on pre-built models.

95) What is the difference between UserPrincipalName (UPN) and UserName in Power BI?

- **UPN:** Email-like identifier used for authentication and Row-Level Security (RLS). Ensures secure access.
- **UserName:** Display name shown in reports. Primarily for user identification, not security.

96) How can clients modify visualizations in a shared Power BI report?

- **Personalize Visuals** allows users to change visual types, fields, and filters in Power BI Service.
- Users need **edit permissions** for deeper modifications.

97) What are the significant updates in the 2024 version of Power BI?

- **AI-powered visuals** (Smart Narratives, Q&A improvements).
- **Enhanced formatting** and performance optimizations.
- **New model view and layout options** for better report design.

98) Can you schedule a monthly report refresh in Power BI?

Yes, scheduled refreshes can be set **daily, weekly, or monthly** in Power BI Service

under **dataset settings**.

99) What are the file formats available for saving a Power BI file?

- **PBIX:** Standard report file (contains data & visuals).
- **PBIT:** Template file (only report structure, no data).
- **PBIP:** New project format for team collaboration.

100) What is incremental refresh, and how would you implement it in your model?

- **Incremental Refresh** updates only new/modified records instead of reloading the entire dataset.
- **Implementation:**
 1. Create **start and end date parameters** in Power BI.
 2. Use these parameters to filter data in **Power Query**.
 3. Enable **Incremental Refresh** in Data Load settings.
 4. Publish the dataset and configure refresh settings in **Power BI Service**.

101) What are Relationship Modifiers?

- **Cross Filter Direction:** Single or bi-directional filtering between tables.
- **Active/Inactive:** Only one relationship can be active at a time.
- **Cardinality:** Defines relationship types (one-to-one, one-to-many, many-to-many).

102) What is a Factless Fact table?

- A table with **no numerical measures**, only relationships between dimensions.
- **Example:** A student attendance log recording presence but without numeric values.

103) Can you provide examples of Slowly Changing Dimensions (SCD)?

- **SCD Type 1:** Overwrites old data (e.g., updating a customer's address).
- **SCD Type 2:** Stores historical versions as new rows (e.g., tracking address changes).
- **SCD Type 3:** Keeps both current and previous values in the same row.

104) Explain the Decomposition Tree visual and its usefulness.

- A **hierarchical visual** that breaks down measures (e.g., sales) across dimensions (e.g., region, product).
- Helps with **root-cause analysis** by dynamically drilling into data.

105) What are the limitations of using DirectQuery mode?

- **Performance issues** due to real-time querying.
- Limited **DAX functions** (e.g., time intelligence).
- Max **1 million rows** per query.
- **Aggregation and complex calculations** are restricted.

106) Deployment Pipeline in Power BI:

Deployment Pipelines facilitate the development lifecycle by allowing staged deployments (Dev, Test, Prod) in Power BI Service: Create stages (Development, Test, Production) within the pipeline.

Publish and manage versions of reports, datasets, and dashboards. Move content between stages, ensuring a controlled and consistent deployment process. Automate testing and approval flows to ensure data integrity.

107) Explain the concept of bidirectional cross-filtering in Power BI. When would you use it, and what are its potential drawbacks?

Bidirectional cross-filtering allows filters to flow in both directions between two related tables (as opposed to the default one-way filter from one table to another).

When to use it:

It is useful in scenarios like many-to-many relationships or when you want users to filter from both sides of a relationship (e.g., both Customers and Orders tables affecting each other).

Drawbacks:

It can lead to ambiguous relationships and circular references, resulting in errors or performance issues. It may create unintended filtering behavior, making reports harder to troubleshoot. More processing is required because both tables need to be checked for filters.

108). What are the best practices for version control and collaboration when working on Power BI projects in a team environment?

Power BI Deployment Pipelines: Use deployment pipelines to manage the lifecycle of reports across development, testing, and production environments.

Source Control: Save Power BI files in version-controlled repositories like Git or use OneDrive/SharePoint for collaboration. Use .pbix files for versioning or tools like Tabular Editor for working with metadata.

Workspaces: Collaborate using Workspaces in Power BI Service to manage shared datasets and reports.

Naming Conventions and Documentation: Maintain consistent naming conventions for datasets, measures, and fields to ensure readability and ease of collaboration.

Use shared datasets: Encourage sharing reusable datasets rather than duplicating work for consistency and performance benefits.

109). Can we pin visualization 1 from Report 2, which is available in workspace 2, into Dashboard 1 (where visualization 1 from report 1 is already pinned)?

No, you cannot pin visualizations from different workspaces into the same dashboard. Power BI dashboards are workspace-specific, meaning that visualizations pinned to a dashboard must all come from reports that exist within the same workspace. Since Report 1 is in Workspace 1 and Report 2 is in Workspace 2, you cannot pin a visualization from Report 2 into Dashboard 1.

Workaround:

You can recreate or duplicate reports into a single workspace, then pin visuals to the same dashboard.

Another option is to share insights between reports by using shared datasets or linked reports within the same workspace, but dashboards themselves cannot span across workspaces.

110). There is a "Month" column, and the months on the X-axis are not sorted correctly. What could be the possible reason for this?

The "Month" column might be stored as a text data type, which causes incorrect sorting. Converting it to a date or numeric type, or setting a custom sort order, would fix the issue.

111). While importing data from SQL with millions of rows and a heavy table, what techniques did you use to reduce the load in Power BI?

I've used methods like query folding, incremental refresh, and partitioning large tables to optimize the import process.

112). Can we use measures in a slicer?

No, slicers in Power BI only work with columns. Measures can't be directly used in slicers, instead we can use field parameters

113).What custom visuals have you used?

I've used custom visuals like bullet charts, KPI indicators, and various other visuals from the Power BI marketplace to enhance data representation.

114) What are the prerequisites for establishing a data connection in Power BI?

To establish a connection, you need access permissions, user credentials, the connection string or data source URL, and any necessary drivers installed.

115) How do you convert categorical data to continuous data in Power BI?

Use DAX functions like IF or SWITCH for a continuous numerical scale, or use grouping and binning features in Power BI's modeling tools.

116) Can you provide a use case of bookmarks from one of your projects in Power BI?

In my HR project, I used bookmarks to create interactive navigation between "Monthly Trends" and "Employee Breakdown" views, making it easier for users to explore different perspectives without losing context.

117) What should be done if the report size exceeds 1GB in Power BI under a Pro license?

Optimize the data model by removing unnecessary columns, refining DAX calculations, or aggregating data. Alternatively, upgrade to Power BI Premium for larger data capacity.

118) Why would you use a Date Table in Power BI if there is already a date column in the dataset?

A dedicated Date Table enables full use of time intelligence functions like YTD, QTD, and MOM, which may not work with simple date columns.

119) How would you calculate the last 3 months of sales data based on the country in Power BI?

`Sales_Last3Months = CALCULATE([Total Sales], DATEADD(Date[Date], -3, MONTH))`

120) Can you provide a use case for using a Live Connection in Power BI?

In a customer segmentation project, I used a Live Connection to SQL Server Analysis Services for real-time data, enabling decision-makers to view up-to-date metrics without a manual refresh.

121) What techniques can be used to reduce the file size of a Power BI report?

Techniques include reducing the number of visuals, disabling unnecessary auto date/time hierarchies, aggregating data, and using Direct Query instead of Import mode for large datasets.

122) Describe the different layers involved in Power BI architecture.

The three layers are:

- **Data Integration layer:** Loading and transforming data
- **Data Modeling layer:** Relationships and measures
- **Visualization layer:** Dashboards and reports

123) How do you implement data lineage in Power BI?

Use the Power BI service to visualize data lineage through the Data Lineage view, which shows dependencies between datasets, reports, and dataflows.

124) What is the role of the M language in Power BI?

M is used in Power Query for data transformation, enabling users to clean, reshape, and combine data before loading it into the model.

125) How do you create a gauge chart in Power BI?

Add a Gauge visual, set the value field (e.g., sales), and define target and range values to represent progress visually.

126) What is the difference between a heat map and a filled map in Power BI?

A heat map uses color intensity to show data density, while a filled map colors geographical areas based on data values.

127) Explain the concept of data masking in Power BI and its use cases.

Data masking hides sensitive information using techniques like obfuscation or masking patterns to maintain confidentiality. It's useful for compliance and security.

128) What is the function of the "Append Queries" feature in Power BI, and how is it used?

"Append Queries" combines rows from two or more tables with the same structure into one table, useful for consolidating data sources.

129) How can you ensure that Power BI recognizes a specific column as a date column if it doesn't do so automatically?

Change the column's data type to "Date" in Power Query or the Data View, and ensure proper date formatting in the source.

130) Describe the process Power BI uses to handle large datasets exceeding the in-memory capacity.

Power BI uses Direct Query or Composite Models to query data directly from the source without loading it entirely into memory.

131) What is the purpose of the VertiPaq engine in Power BI?

The VertiPaq engine compresses data and performs in-memory storage for fast query execution and efficient performance with large datasets.

132). How can you handle time zone conversions in Power BI?

Use DAX functions like UTCNOW(), adjust time zones using calculated columns or measures, or handle conversions in Power Query with custom transformations.

133). How do you use R or Python scripts in Power BI?

R/Python scripts can enhance Power BI by enabling advanced analytics. For instance, I used Python to apply clustering (K-means) for customer segmentation and visualized the clusters in Power BI.

134). Write a DAX measure to calculate the year-over-year growth in revenue for each month.

YoY Growth =

VAR CurrentMonthRevenue = SUM(Sales[Revenue])

VAR LastYearRevenue = CALCULATE(SUM(Sales[Revenue]),
SAMEPERIODLASTYEAR(Sales[Date]))

RETURN

IF(LastYearRevenue = 0, BLANK(), (CurrentMonthRevenue - LastYearRevenue) /
LastYearRevenue)

135). Write a DAX formula to calculate cumulative sales for each product across the entire sales period.

Cumulative Sales =

CALCULATE(

SUM(Sales[SalesAmount]),

FILTER(

ALLSELECTED(Sales[Date]),

Sales[Date] <= MAX(Sales[Date])

)

)

136) DATESBETWEEN vs. DATESINPERIOD

DATESBETWEEN: Returns dates between two specific dates.

DATESINPERIOD: Returns dates within a specified period (e.g., last 30 days) relative to a date.

137). Did you have access to a premium Power BI account?

I used both Pro and Premium features, leveraging Premium for larger datasets and advanced features like AI insights and paginated reports.

138). Difference Between Data Lake, Data Warehouse, and Data Mart?

Data Lake: Raw data storage for exploratory analysis.

Data Warehouse: Structured data for reporting and analytics.

Data Mart: Subset of a Data Warehouse for specific functions like sales.

139). Difference Between Structured and Unstructured Data?

Structured Data: Organized, easy to analyze (e.g., sales logs).

Unstructured Data: No predefined structure, requires tools for analysis (e.g., videos, emails).

140). How do you integrate Power BI with other Microsoft services?

Excel: Import or export data.

SharePoint: Embed reports or fetch data.

Teams: Share and collaborate on reports.

Azure: Connect with Azure SQL, Synapse, or Data Lake.

Power Automate: Automate workflows based on Power BI triggers.

141). How do you set up data alerts in Power BI?

Create a KPI, card, or gauge visual in your report.

Publish the report to Power BI Service.

Set up alerts by clicking the visual's ellipsis (...) and selecting "Manage alerts."

Define conditions for the alert (e.g., threshold values).

Enable email notifications.

142). How does the end user access the report?

Through Power BI Service via shared links or apps.

Embedded in web portals or applications.

Shared via Microsoft Teams or email.

Exported as PDF or PowerPoint presentations.

143).Connecting two fact tables in Power BI:

Use a bridge table or shared dimension.

Avoid direct relationships between fact tables to prevent ambiguity.

Alternatively, use DAX measures for linking.

144).VALUES vs DISTINCT in DAX

Both VALUES and DISTINCT return unique values from a column, but there's a subtle difference in their behavior!

Key Difference:

VALUES() returns a column with distinct values and also includes BLANK() values if present.

DISTINCT() returns only distinct non-blank values from a column.

Use Case:

VALUES() is useful when you want to include a context filter and keep BLANK() values.

DISTINCT() is ideal when you need distinct values without blanks.

145).What is the significance of specifying 0 as the third parameter in the DIVIDE() function in DAX?

It acts as a default value to avoid division by zero errors, ensuring calculations remain error-free.

146). If data flow from Table A to Table B is already defined, how would you define the reverse data flow from Table B to Table A?

Create a bidirectional relationship in Power BI to allow data propagation both ways. Alternatively, adjust query logic in the ETL process for reverse flow.

147) Explain Row Context vs. Filter Context in DAX with an example.

Row Context: Applies when a calculation is evaluated row by row. For example, using SUMX iterates over each row in a table and computes a result.

Filter Context: Refers to the filters applied to the data model in a report. For example, when you use a slicer, you apply a filter context that modifies the calculation based on the filtered data.

148). How do you handle circular relationships in Power BI?

Circular relationships occur when there is ambiguity in relationships between tables, creating a loop.

Remove unnecessary relationships causing circular dependency.

Use one-to-many (1:M) relationships instead of many-to-many (M:M) where possible.

Use inactive relationships with USERELATIONSHIP() in DAX to activate relationships dynamically.

Consider using bridge tables to resolve complex many-to-many relationships.

149). How can you change the order of values on the X-axis of a Column Chart?

You can change the order in two ways:

Sort By Column: Select the category field in the Fields pane → Click "Sort By Column" → Choose a numerical column that defines the custom order.

150). Your PBIX file is not getting published—what could be the reasons?

Possible reasons include:

File size exceeds 1GB (for Pro users) or 10GB (for Premium).

Limited workspace storage, requiring cleanup of unused datasets.

Dataset contains unsupported DirectQuery connections to certain data sources.

Power BI Service restrictions, such as license limitations.

Network issues or authentication errors during publishing.

151. How do you embed Power BI reports into a web application?

Use Power BI Embedded (Azure) for authentication-based embedding.

Generate an embed token using the REST API.

Embed reports using the JavaScript API (Power BI Client SDK).

152. What is Cardinality & its Impact on Relationships?

Cardinality defines the uniqueness of values in a column when creating relationships. Power BI supports one-to-one (1:1), one-to-many (1:M), and many-to-many (M:M) relationships. Incorrect cardinality can cause duplication, incorrect aggregations, and performance issues. Always analyze data before defining relationships.

153. How to Optimize Power BI Data Models?

Use star schema instead of snowflake.

Remove unnecessary columns & rows.

Optimize DAX by avoiding nested iterators.

Use Aggregations, Summarized tables, and Query Folding for efficiency.

154. How do FILTER and ALL functions work in DAX?

FILTER returns a subset of data based on conditions.

ALL removes all filters from a table/column.

Total Sales = SUMX(FILTER(Sales, Sales[Category] = "Electronics"), Sales[Amount])

Total Sales (Ignoring Filters) = CALCULATE(SUM(Sales[Amount]), ALL(Sales))

155. What is the use of the EARLIER function in DAX?

EARLIER is used in row context to refer to a previous row's value in the same column. It's commonly used in calculated columns and ranking scenarios.

Rank = RANKX(FILTER(Sales, Sales[Category] = EARLIER(Sales[Category])), Sales[Amount])

156. How do you calculate running totals in Power BI using DAX?

Use the CALCULATE function with FILTER on dates:

Running Total =

```
CALCULATE(  
SUM(Sales[Amount]),  
FILTER(  
ALL(Sales[Date]),  
Sales[Date] <= MAX(Sales[Date])  
)  
)
```

157. How do you automate Power BI report refreshes?

Scheduled Refresh in Power BI Service.

Power Automate for workflow-based automation.

REST API for advanced refresh automation.

158. How do you use Python or R inside Power BI for advanced analytics?

Use the Python/R visual to write custom scripts.

Perform data transformations or build machine learning models using Pandas, NumPy, or Scikit-learn.

Example: Running a regression model inside Power BI using Python.

159. What is the role of Power BI REST API?

Power BI REST API is used for:

Automating dataset refreshes

Embedding reports in applications

Managing users, groups, and workspaces

Fetching dashboard metadata programmatically

160. What are the different types of joins available in Power Query?

Power Query provides inner join, left outer join, right outer join, full outer join, anti join, and cross join. Inner join returns only matching records, while outer joins include unmatched records from one or both tables. Anti join filters out matching records, and cross join creates all possible combinations of rows between tables.

161. How do you create a star schema in Power BI?

A star schema consists of a central fact table connected to multiple dimension tables. First, define the fact table with numeric measures and create dimension tables with descriptive attributes. Establish one-to-many relationships between dimension and fact tables to improve query performance and maintain data integrity.

162. What is the importance of surrogate keys in data modeling?

Surrogate keys are system-generated unique identifiers used in dimension tables instead of natural keys. They ensure consistency, improve performance by reducing join complexity, and help maintain historical data integrity, especially in slowly changing dimensions.

163. How does Power BI handle slowly changing dimensions SCDs?

Power BI can handle SCDs using Power Query transformations, calculated columns, or DAX logic. Type 1 updates records by overwriting old data, Type 2 maintains historical records with versioning, and Type 3 keeps limited history within the same row using additional columns.

164. What are composite models in Power BI, and how do they work?

Composite models allow a mix of Import and DirectQuery data sources in the same report. Frequently used data can be stored in Import mode for performance, while real-time analysis is done using DirectQuery. This provides flexibility in handling large datasets without overloading memory.

165. What is the difference between ADDCOLUMNS and SUMMARIZE in DAX?

ADDCOLUMNS adds calculated columns to a table without aggregation, whereas SUMMARIZE groups data before adding columns. SUMMARIZE is useful for creating aggregated summary tables, while ADDCOLUMNS is better for adding row-level calculations to an existing table.

166. How do you calculate weighted averages in DAX?

A weighted average is calculated using SUMX to multiply values by their respective

weights and then dividing by the total weight. This method ensures each value contributes proportionally based on its weight, commonly used in financial and sales analysis.

167.What is the purpose of GENERATE and GENERATESERIES in DAX?

GENERATE creates a cross-join between two tables, expanding one table based on another. GENERATESERIES creates a continuous sequence of numbers or dates, useful for generating missing time periods or custom axes in charts.

168.How do you create a moving average calculation in Power BI?

A moving average is calculated using AVERAGEX with DATESINPERIOD to dynamically compute the average over a rolling time frame. This technique helps smooth out short-term fluctuations in trends by considering past periods.

169.What is the difference between SUMX and CALCULATE in DAX?

SUMX iterates row by row, performing calculations before aggregating, making it useful for weighted measures. CALCULATE modifies filter context to apply conditions dynamically, commonly used for conditional aggregations like year-over-year comparisons.

170.What are the best practices for optimizing DAX queries?

Use variables to store intermediate calculations, avoid nested iterators, and prefer measures over calculated columns. Minimize row-based operations, optimize relationships, and leverage query folding where possible for better performance.

171.How do you identify and fix performance bottlenecks in Power BI?

Use DAX Studio to analyze query performance, reduce dataset size by removing unnecessary columns, and optimize relationships. Aggregations, pre-calculated tables, and indexed columns in the data source can also improve speed.

172.How does Query Folding impact Power BI performance?

Query folding pushes transformations back to the data source, reducing processing in Power BI. It improves refresh performance and is essential for handling large datasets efficiently, especially in DirectQuery mode.

173.What are the advantages of using aggregations in Power BI?

Aggregations improve performance by pre-calculating summary data, reducing the number of records processed at runtime. This speeds up queries and is beneficial when working with large datasets where detailed-level queries are not always required.

174.How can you reduce the size of a Power BI dataset?

Remove unnecessary columns, use integer keys for relationships, optimize calculated

columns, and prefer Import mode only for essential tables. Aggregating data, reducing granularity, and disabling auto date/time tables can further optimize the model.

175.What are the limitations of scheduled refresh in Power BI Service?

Scheduled refresh is limited to eight times per day for Power BI Pro and 48 times for Premium. Large datasets may experience memory constraints, and on-premises sources require a configured gateway to support refresh.

176.How do you troubleshoot gateway connectivity issues?

Check if the gateway is online, verify credentials, and ensure firewall settings allow Power BI access. Restart the gateway service, match data source settings with Power BI Service, and update the gateway to the latest version.

177.What is incremental refresh, and how is it different from a full refresh?

Incremental refresh updates only new or modified records instead of reloading the entire dataset, improving performance. A full refresh reloads all data, which can be time-consuming for large datasets and lead to slower processing.

178.What are the security best practices for publishing reports in Power BI Service?

Implement row-level security to restrict data access, limit sharing permissions, disable export options, and configure workspace security roles. Use Microsoft Purview to classify sensitive data and enforce compliance policies.

179.How do you configure data source credentials for cloud and on-premises sources?

Cloud sources use OAuth, API keys, or basic authentication, while on-premises sources require a data gateway. Credentials must be updated in Power BI Service under dataset settings to ensure a smooth refresh process.

180.How do you create a waterfall chart in Power BI?

A waterfall chart shows changes in a value over time by using a column chart format with intermediate increases and decreases. It is used for financial and trend analysis, helping to track the impact of different factors on a total value.

181.How do you implement drill-through in Power BI reports?

Drill-through allows users to navigate from a summary page to a detailed report by right-clicking a visual. A separate drill-through page is created with the relevant field added to the filters pane for interactivity.

182.What is the difference between a slicer and a filter in Power BI?

A slicer is a visual element that allows users to interactively filter data, while a filter is

applied in the background without a visible component. Slicers offer better user control, whereas filters provide more flexibility in defining report-wide conditions.

183.How do you create a dynamic axis in a Power BI visual?

Use field parameters to allow users to switch between different axis fields dynamically. This enhances interactivity by enabling users to analyze data across multiple dimensions.

184.How do you create small multiples in Power BI?

Small multiples are created using the small multiples feature in charts, allowing the same visual to be displayed across different categories. This is useful for comparing trends across multiple segments.

185.How does Power BI handle real-time streaming datasets?

Power BI can connect to streaming data sources using push datasets, Azure Stream Analytics, or PubNub. Reports update in real-time without requiring manual refresh, enabling live monitoring of metrics.

186.What is the difference between paginated reports and regular Power BI reports?

Paginated reports are designed for printing with a fixed layout, whereas regular Power BI reports are interactive and optimized for on-screen visualization. Paginated reports are commonly used for financial statements and invoices.

187.How do you integrate Power BI with Power Automate?

Power Automate can trigger workflows based on Power BI events, such as sending email alerts, refreshing datasets, or exporting reports. This helps automate repetitive tasks and improve report distribution.

188.What are Power BI goals, and how do they help track performance?

Power BI goals allow users to set and monitor key performance indicators in Power BI Service. They help in tracking business objectives with real-time updates and historical performance trends.

189.How do you use AI visuals like Key Influencers in Power BI?

The Key Influencers visual uses machine learning to analyze data and identify key factors affecting an outcome. It helps in understanding what drives specific trends or behaviors in the dataset.

190How do you enable and use sensitivity labels in Power BI?

Sensitivity labels classify data based on confidentiality levels and restrict sharing,

exporting, and printing. They help organizations enforce data security and compliance policies across Power BI reports and datasets.

BONUS: 10 Behavioral Questions and Answers:

191. Tell me about a time when you had to work under a tight deadline.

There was a time when I had just two days to build a Power BI dashboard for senior management. I quickly identified the key metrics they needed, streamlined the data model, and focused on essential visualizations. I also kept them updated on progress so there were no surprises. In the end, I delivered a functional report on time, and they appreciated the clarity and quick turnaround.

192. Describe a situation where you faced a challenge while working with stakeholders.

A client once insisted on real-time data in a Power BI report, but their system only supported daily updates. Instead of just saying no, I explained the technical limitations in simple terms and suggested a hybrid solution—real-time data for key metrics via DirectQuery and scheduled refreshes for everything else. This way, their critical insights were always up-to-date, and they were happy with the compromise.

193. How do you handle multiple projects with competing deadlines?

I had a situation where I was juggling multiple reports for different teams, all with tight deadlines. I started by listing out priorities, setting clear timelines, and breaking tasks into manageable steps. I also communicated early with stakeholders to set realistic expectations. By staying organized and focusing on one task at a time, I was able to deliver everything without last-minute panic.

194. Give an example of a time when you made a mistake in your work and how you handled it.

Once, I deployed a Power BI report with a filtering issue that caused incorrect data to be shown. I realized it after someone pointed it out, and instead of panicking, I immediately investigated, fixed the problem, and sent an update to the team explaining the correction. After that, I introduced a simple checklist for myself to double-check filters before publishing reports. It was a good learning experience.

195. Have you ever had to convince a team or manager to adopt a new tool or approach?

Yes, my team used to manually generate Excel reports, which took hours every week. I

built a Power BI dashboard as a proof of concept to show how automation could save time. At first, some were hesitant to switch, but once they saw how easy it was to use and how much time it saved, they fully embraced it. Eventually, we automated multiple reports and cut reporting time in half.

196. How do you handle feedback and criticism?

I try to take feedback as a way to improve rather than taking it personally. In one project, I designed a dashboard that I thought was great, but my manager said it felt too cluttered. Instead of getting defensive, I asked what specifically could be improved. Based on their feedback, I simplified the visuals, and the final version was much better. Now, I always ask for feedback earlier in the process.

197. Tell me about a time you had to explain a complex concept to a non-technical audience.

A manager once asked me why Power BI was running slow, and I realized they didn't understand the difference between DirectQuery and Import mode. Instead of using technical jargon, I compared it to ordering food—DirectQuery is like ordering fresh every time, and Import mode is like having a buffet ready. That simple explanation made them understand why certain queries took longer, and they appreciated the clarity.

198. Describe a situation where you had to quickly learn a new tool or skill.

I was once given a task that required writing SQL queries, but I wasn't very experienced with it. Instead of stressing, I spent the weekend going through online tutorials and practicing queries with sample data. By the time I had to work on the project, I was confident enough to write optimized queries. It was a great reminder that learning something new quickly is always possible with the right mindset.

199. How do you deal with a difficult teammate?

I once worked with a teammate who missed deadlines and was hard to reach. Instead of assuming they were careless, I casually checked in and asked if they needed help. It turned out they were overwhelmed with multiple tasks. We split the work more evenly, and after that, our collaboration improved a lot. Sometimes, a little understanding goes a long way.

200. Tell me about a time you went above and beyond to solve a problem.

Our team had a recurring issue where a report refresh would fail unpredictably, delaying decisions. I took the initiative to investigate beyond just fixing it for the moment—I analyzed query performance, optimized transformations, and implemented incremental refresh. This cut down processing time by 60%, and the issue never

happened again. My manager really appreciated that I didn't just fix the symptom but solved the root cause.